

Annex

List of python files done in notebook

API_ingestion.ipynb (not workable as there is no real API, just as per requirements mock up solution)

Files: Silver_Final.ipynb and Gold_Final.ipynb

The code is still in progress and just represents basic processing.

1. Folder structure to hold data

```
your_project_folder/  
  Bronze/  
  Silver/  
  Gold/
```

Bronze folder

- Put your input original files here
- Files must be named like:

Agents Team List YYYYMMDD.xlsx

Telephony YYYYMMDD.csv

Schedules YYYYMMDD.csv

2. Transformation

Silver_Final.ipynb saves transformed files into silver folder

Procedures:

```
process_agent_team_list("path_to_your_project_folder") #First to be run
```

What the code does

- Reads team list Excel files from Bronze
- Cleans and standardises dates
- Handles duplicate records for the same employee
- Adjusts date ranges so they do not overlap
- Creates a unique key for each employee record
- Generates a surrogate key (sk_employee)
- Saves cleaned data to Silver

Key things to check

- File naming must be correct
- Required columns must exist
- Dates should be valid (invalid ones become empty)
- Empty employee_id is not yet handled (noted in code)

Output Silver/ Schedules_YYYYMMDD.csv

```
process_telephony_pipeline("path_to_your_project_folder")
```

Silver folder (required BEFORE running)

Must already contain: dim_clean_YYYYMMDD.csv

What the code does

- Reads telephony files from Bronze

- Cleans basic issues (dates, missing values)
- Matches agents to dimension table
- Assigns surrogate keys (sk_employee)
- Saves cleaned data to Silver

Key things to check

- File naming must be correct (Telephony_YYYYMMDD.csv)
- Dimension file must exist in Silver
- Agent names must match (otherwise no key assigned)
- Invalid dates will be flagged

Output Silver/ Telephony_YYYYMMDD.csv

`process_schedules_pipeline("path_to_your_project_folder")`

Silver folder (required BEFORE running)

- Must already contain: dim_clean_YYYYMMDD.csv

What the code does

- Reads schedule files from Bronze
- Cleans invalid or missing times
- Fixes overlapping time records (ensures no overlaps per employee per day)
- Matches employees to dimension table
- Assigns surrogate keys (sk_employee)
- Saves cleaned data to Silver

Key things to check

- File naming must be correct (Schedules_YYYYMMDD.csv)
- Dimension file must exist in Silver
- Start and end times must be valid
- Overlapping records will be adjusted (not removed)

Output Silver/ Schedules_YYYYMMDD.csv

3. Business ready layer (gold)

(Deduplication of Telephony, concatenation to history, get the latest team list)

`update_agent_manager_dim("path_to_your_project_folder")`

What the code does

- Finds the latest cleaned dimension file in Silver
- Deletes any existing output file in Gold
- Copies the latest file into Gold
- Renames it to a standard name

Key things to check

- Silver folder must contain at least one dim_clean file
- Gold folder must exist
- Existing file in Gold will be **deleted and replaced**

Output Gold/ agent_manager_dim.csv

`update_gold_schedules("path_to_your_project_folder")`

What the code does

- Reads all schedule files from Silver
- Combines them into one dataset
- Reads existing Gold file (if it exists)
- Appends new data to existing data
- Saves updated dataset to Gold

Key things to check

- Silver folder must contain schedule files
- Gold folder must exist
- Existing Gold file will be **updated (not replaced)**
- Empty or broken files are handled safely

Output Gold/ Schedules.csv

update_gold_telephony("path_to_your_project_folder")

What the code does

- Reads all telephony files from Silver
- Combines them into one dataset
- Removes invalid records (wrong date or missing agent)
- Keeps only the latest record per employee and date (deduplicate)
- Removes temporary flag columns
- Reads existing Gold file (if it exists)
- Appends new cleaned data
- Saves final dataset to Gold

Key things to check

- Silver folder must contain telephony files
- Gold folder must exist
- Files must include required columns
- Existing Gold file will be **updated (not replaced)**
- Deduplication is based on latest load_timestamp

Output Gold/ Telephony.csv